Allan Wang

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EDUCATION

Bachelor of Science, Combined Major in Computer Science and Statistics Sept 2022 - May 2027
University of British Columbia – Cumulative GPA: **4.33/4.33** Vancouver, BC
Relevant courses: Relational Databases, Computer Systems, Algorithms & Data Structures, Statistical Modelling

TECHNICAL SKILLS

Programming: C, C++, JavaScript, TypeScript, Java, Python, Swift, SQL, R, HTML, CSS
Technologies: React.js, Node.js, Next.js, PostgreSQL, Java Swing, Python pandas, Spring Boot, RTOS, OpenThread, SFML, Oracle Database, IoT, Express.js, Linux, Supabase
Developer Tools: Git, Repo, Postman, Docker, Vercel, JUnit, GoogleTest, Mocha

EXPERIENCES

Software Engineer Intern – Infotainment Platforms Sep 2025 – Apr 2026
Rivian and Volkswagen Group Technologies Vancouver, BC

- Developed infotainment platform service for centralized performance metrics monitoring; processed metrics into summary KPIs and reported to telemetry across multiple Virtual Machines with C++.
- Implemented proc-based system-wide and per-process CPU, memory, and I/O scanners, reducing metric collection cycle latency by **96% (15s → 0.6s),** bypassing delays caused by vendor API limitations.
- Refactored scanners via a registry singleton and base scanner strategy, reducing ~200 lines of code/scanner.
- Designed unit tests for scanners, service overhead tests, and validation tests to ensure service outputs matched built-in Linux command measurements within a **2%** deviation.
- Resolved SELinux denial flooding in logcat by configuring read permissions, reducing **>20 entries/second**.

Software Engineer Intern – R&D Dec 2024 – Aug 2025
LED Smart Inc. Surrey, BC

- Designed and implemented embedded software on nRF54L15 using Zephyr RTOS and OpenThread in **C**; leveraged peripherals such as **GPIO and UART** to achieve control over LED states as a robust Thread network.
- Refined Thread device commissioning process by applying mutex locking and isolated blocking functions with multi-threading, reducing main thread delay by **1500ms** and improving system responsiveness during boot-up.
- Implemented and validated custom networking protocols using **UDP** with custom reliability mechanisms.
- Responded actively to QA testing feedback to advance the project from initial development to **beta release**.

Full-Stack Developer Jan 2025 – Aug 2025
UBC Cubing (Rubik's Cubes) Club Vancouver, BC

- Designed, developed and maintained the website for the UBC Cubing Club to be used for weekly meetings.
- Created a custom timer with reliable client-side submissions; this increased per-member solve capacity from 5 to 40 per meeting (~**8× increase**), **cut printing costs & effort entirely**, and boosted average attendance by **200%**.
- Designed tables in PostgreSQL database with Supabase and enforced Row Level Security for data integrity.

Coding Instructor Aug 2024 – Nov 2024
Robokids Canada White Rock, BC

- Assisted students with LeetCode-style Python & Java questions; reviewed work of students and tested them for bugs and provided insights for optimization and improvements. Led camps and maintained safe environment.

PROJECTS

Pet Management System | Full Stack, JavaScript (React, Node, Express), SQL, Database Jan 2025 – Apr 2025

- Developed a web application using **React.js with JavaScript**, that allows users to perform **CRUD** operations on a pet shelter database enabled using **custom APIs** with structured routing.
- Designed, normalized, and implemented an Oracle Database schema to optimize data storage and efficiency.

Algorithm Visualizer | Swift, SwiftUI, Algorithms, iOS Development Aug 2024 – Dec 2024

- Developed an algorithm visualizer that solves customizable mazes as an iOS application with Swift & SwiftUI.
- Implemented **BFS and DFS** to solve the maze designed by the user, rendered the solution as stepped animations.